

Behind the Stars

Mining Customer Reviews for Restaurant Hygiene Using Fuzzy Logic

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Abstract

- Star ratings are commonly used to evaluate restaurants, but they often fail to reflect hygiene and cleanliness conditions.
- Customers may give high ratings for good food or service while mentioning hygiene problems only in written reviews. In this project, we analyze Google Maps review text and apply fuzzy logic to convert subjective hygiene-related opinions into a clear hygiene score for each restaurant.
- Our visual analysis shows that star ratings do not always represent hygiene quality, helping reveal potential risks hidden behind high ratings.

Problem Statement

- Hygiene and cleanliness are critical factors for customer health and safety, yet they are not clearly represented by star ratings.
- Many customers focus on food quality and service when rating restaurants, while hygiene concerns are mentioned only in review text.
- This can cause important warning signs, such as bad smells or dirty environments, to be overlooked. By highlighting these hidden hygiene signals, we aim to provide a clearer and more realistic view of restaurant conditions.

$$\text{Hygiene Sentiment} = \frac{N_{pos} - N_{neg}}{N_{pos} + N_{neg}}$$

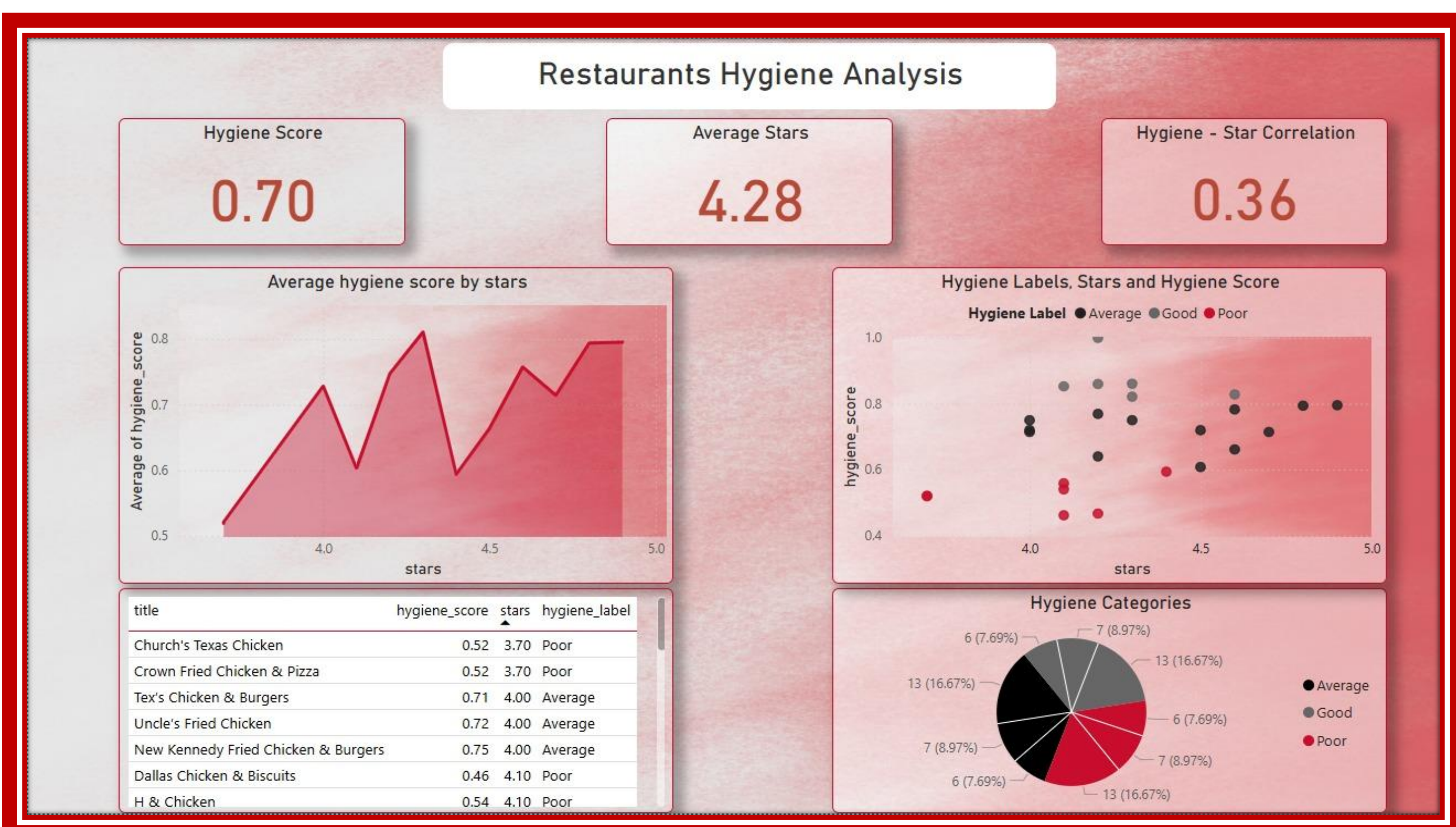
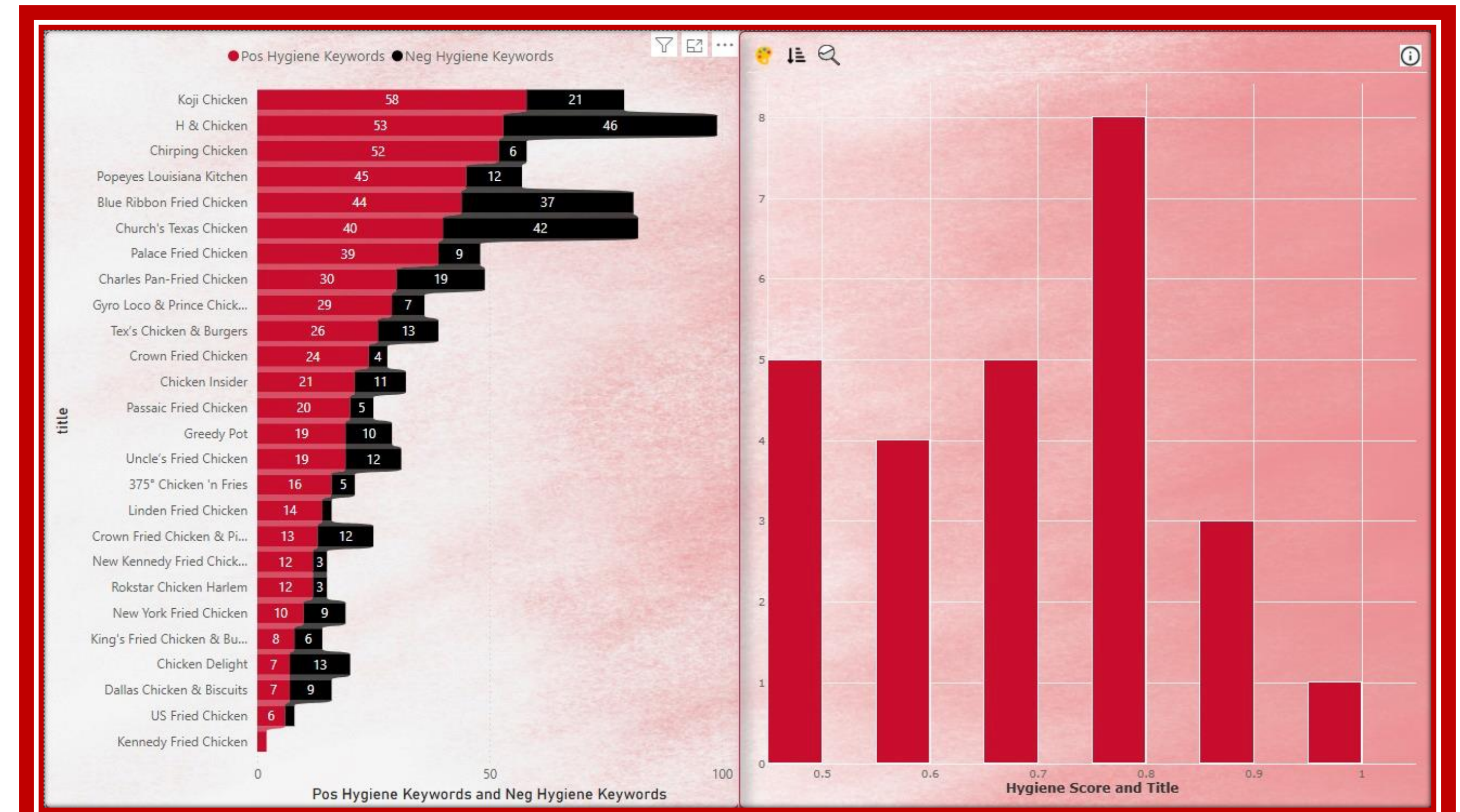
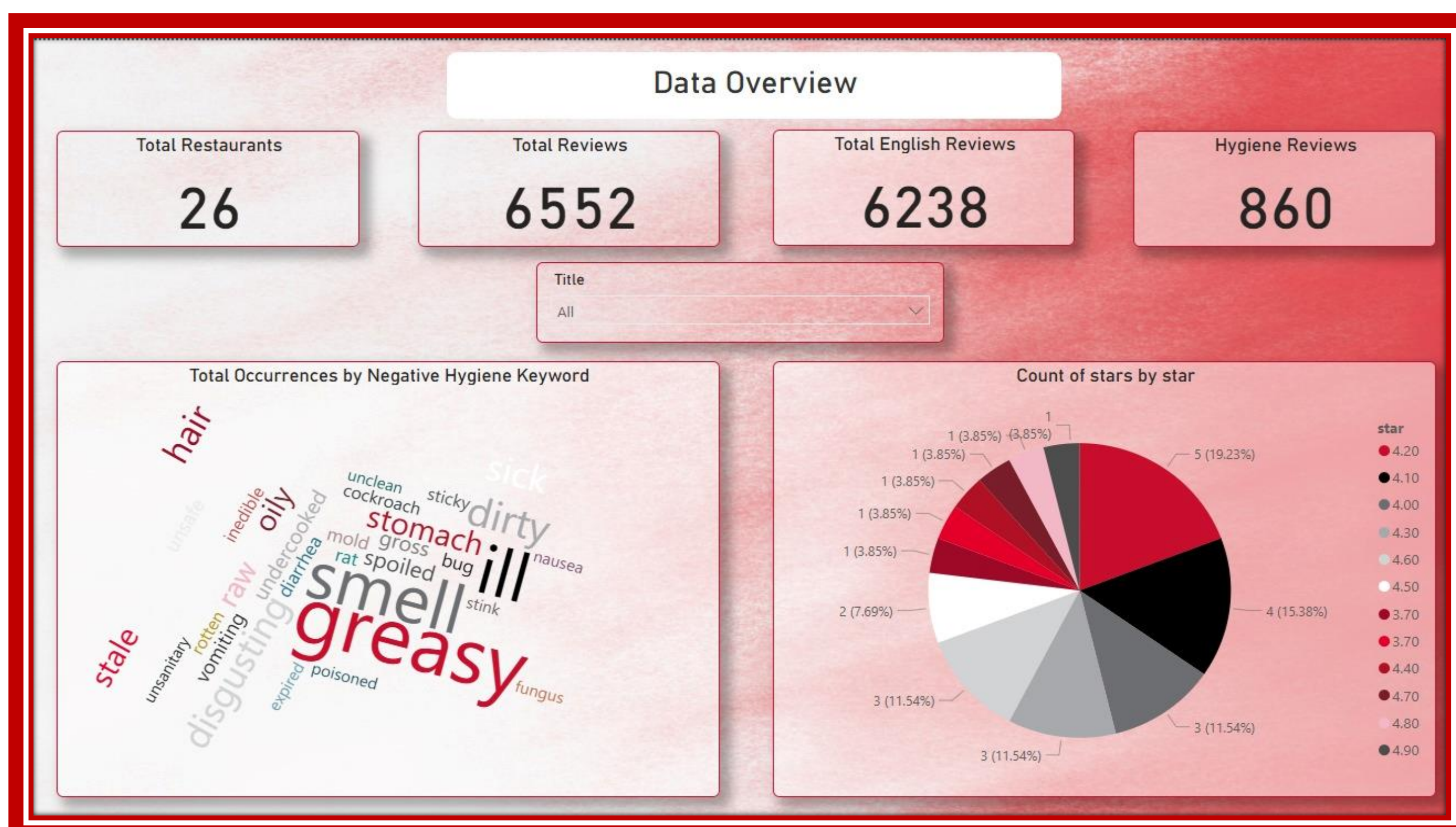
Introduction

- Online star ratings play a major role in how people choose restaurants and form expectations about quality.
- However, these ratings usually reflect factors such as taste, service, and price rather than hygiene and cleanliness.
- Customers often mention hygiene issues only briefly in their written reviews, which makes this information easy to miss.
- As a result, restaurants with high ratings may still have cleanliness problems that are not immediately visible to users.

System workflow



Results



Conclusion

- This project shows that star ratings alone are not reliable indicators of restaurant hygiene. By analyzing customer review text and applying fuzzy logic, we were able to uncover hygiene signals that are often hidden behind high ratings. The results reveal clear mismatches between stars and cleanliness, highlighting restaurants that may appear highly rated but still raise hygiene concerns. This approach demonstrates how text mining and fuzzy systems can provide a more realistic and interpretable assessment of restaurant hygiene.

References

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